

ASTRAKHAN, Russian Federation

WMO: 348800

Lat: 46.283N

Lon: 47.983E

Elev: -68

StdP: 14.73

Time Zone: 4.00 (E04)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-1.4	4.8	-7.5	3.7	0.2	-1.5	5.1	5.9	26.5	23.5	23.3	26.8	6.5	60	0.549

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	21.9	98.4	69.4	95.2	68.8	92.7	68.2	74.3	86.5	72.6	85.6	71.2	84.8	10.6	110

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
70.9	113.6	79.3	68.5	104.5	77.7	66.7	98.1	76.9	37.7	85.9	36.2	85.7	34.9	84.6	80.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
24.1	21.0	18.5	-5.9	101.9	9.5	3.8	-12.7	104.6	-18.2	106.8	-23.6	108.9	-30.4	111.7		
			DB													
			WB	-6.2	76.9	9.2	1.9	-12.8	78.3	-18.2	79.4	-23.4	80.4	-30.1	81.8	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	52.4	25.1	27.3	38.7	52.9	65.7	75.1	79.5	77.5	65.4	52.2	38.5	29.4	
	DBStd	20.92	9.84	10.26	7.28	7.62	7.33	6.32	4.88	6.02	7.12	8.20	9.13	8.87	
	HDD50	2883	771	637	353	55	1	0	0	0	1	73	353	639	
	HDD65	5943	1236	1057	814	369	82	6	0	1	80	400	794	1104	
	CDD50	3769	0	0	4	141	489	754	914	853	462	142	10	0	
	CDD65	1353	0	0	0	5	105	310	449	389	91	4	0	0	
	CDH74	15091	0	0	0	82	1066	3427	5246	4401	826	43	0	0	
	CDH80	7167	0	0	0	11	349	1628	2701	2228	245	5	0	0	
Wind	WSAvg	8.8	8.9	9.3	9.9	9.9	9.1	8.5	7.8	7.9	8.5	8.2	8.3	9.2	
Precipitation	PrecAvg	8.5	0.6	0.5	0.6	0.8	1.0	0.9	0.7	0.7	0.8	0.7	0.7	0.7	
	PrecMax	14.8	1.6	2.0	2.5	3.4	4.4	3.1	3.2	3.1	2.7	3.0	1.7	1.7	
	PrecMin	3.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	
	PrecStd	2.5	0.4	0.4	0.5	0.8	0.9	0.8	0.7	0.7	0.7	0.7	0.4	0.4	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	46.1	50.0	63.0	81.4	91.8	100.4	102.3	102.1	91.6	78.9	62.9	49.8	
		MCWB	43.0	43.3	50.1	60.2	65.3	69.8	70.2	69.0	67.4	62.3	54.1	46.0	
	2%	DB	41.4	45.5	58.2	75.9	87.5	96.0	98.3	98.3	86.1	73.7	57.6	46.0	
		MCWB	39.3	40.5	47.0	57.8	64.4	68.3	69.7	69.3	64.9	58.3	51.7	43.2	
	5%	DB	39.0	42.5	54.0	71.4	84.0	92.9	95.2	94.7	82.4	69.6	53.9	42.7	
		MCWB	37.1	38.7	44.6	55.8	62.9	67.7	69.1	68.8	63.7	56.8	49.7	40.7	
	10%	DB	36.3	39.2	50.4	67.6	80.3	89.3	92.5	91.3	78.8	64.8	50.5	40.0	
		MCWB	34.7	36.5	42.7	53.9	61.9	66.7	68.6	68.3	61.8	55.6	47.2	38.3	
	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	43.3	44.3	51.4	62.1	69.1	75.9	76.4	76.4	71.9	64.5	55.7	46.5
			MCDB	44.9	47.9	62.4	77.7	83.4	88.8	88.1	88.0	83.8	73.2	60.8	48.5
2%		WB	39.7	41.2	48.2	59.4	66.9	72.7	73.8	74.4	68.8	61.5	52.7	43.2	
		MCDB	41.1	44.1	56.2	72.4	80.9	86.2	86.6	86.1	79.4	69.6	56.7	44.7	
5%		WB	37.0	38.9	45.6	57.1	65.3	70.7	72.4	72.5	66.6	58.9	50.5	40.7	
		MCDB	38.4	41.6	52.3	68.9	78.4	84.1	86.0	85.0	77.1	65.6	53.6	42.1	
10%		WB	34.8	36.7	43.4	54.9	63.7	69.1	71.1	71.0	64.5	56.5	47.7	38.3	
		MCDB	36.1	38.8	49.2	64.8	76.0	82.7	85.2	84.4	74.6	63.7	50.5	39.7	
Mean Daily Temperature Range	5% DB	MDBR	11.3	14.3	16.8	20.0	20.3	21.4	21.9	22.8	21.1	18.3	13.3	9.9	
		MCDBR	12.0	17.0	22.5	26.3	25.0	26.5	26.3	27.4	25.1	23.9	17.3	11.9	
		MCWBR	9.8	12.4	13.0	11.4	8.3	8.0	7.5	8.6	9.3	11.2	11.7	9.7	
	5% WB	MCDBR	11.3	15.5	20.7	24.1	21.9	23.3	23.2	24.1	21.9	20.7	15.9	11.2	
		MCWBR	9.6	11.8	12.8	11.3	8.6	8.1	8.0	9.1	9.8	11.7	11.4	9.8	
Clear-Sky Solar Irradiance	taub	0.362	0.390	0.427	0.494	0.462	0.445	0.451	0.456	0.428	0.393	0.371	0.374		
	taud	2.143	2.079	2.063	1.936	2.105	2.206	2.201	2.163	2.183	2.237	2.207	2.111		
	Ebn at Noon	218	238	248	242	257	262	259	251	246	236	215	196		
	Edh at Noon	30	38	45	56	49	44	44	44	39	32	28	28		
All-Sky Solar Radiation	RadAvg	337	617	1056	1559	1964	2156	2081	1862	1418	870	443	256		
	RadStd	51	88	113	136	154	140	105	93	97	90	63	40		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	+1.03	N/A	N/A	+2.70	N/A	N/A	N/A	N/A	+306	+207
(47)	Regional (2 neighbors)	+0.95	N/A	N/A	+2.63	N/A	N/A	N/A	N/A	+306	+207

Nomenclature: See separate page